

VENKATESH MANNURU

Cybersecurity Engineer

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Professional summary

Security Engineer with 7+ years of experience securing large-scale, transaction-heavy platforms across financial and payment systems. Brings a strong, research-driven foundation in CISSP security domains, OSCP-aligned penetration testing methodologies, and Azure cloud security, combined with hands-on expertise in secure software engineering. Proven ability to design and protect .NET-based microservices, RESTful APIs, containerized workloads, and SQL data platforms using secure SDLC, threat modeling, defense-in-depth, and vulnerability remediation. Experienced in safeguarding high-volume payment flows (credit cards, ACH, money orders, cash, and checks), implementing robust authentication, authorization, encryption, auditing, and monitoring, and collaborating cross-functionally to maintain availability, performance, and security at scale. Adept at bridging development and security engineering to deliver resilient, compliant, and cloud-ready systems.

Experience



TransCore

New Jersey, united states

Senior Software Engineer – Application Security

06/2025 - Present

TransCore, a U.S. tolling tech company, offers secure toll collection, payment processing, and transportation infrastructure.

- Designed and developed secure financial applications within a toll-domain platform, supporting high-volume transactions.
- Designed and developed secure financial applications for a toll-domain platform handling high-volume transactions.
- Implemented secure payment processing for credit cards, ACH, money orders, cash, and checks.
- Built secure backend services using C#/.NET and frontend components using Angular, following secure coding practices.
- Developed secure REST APIs for reliable transaction flows. Managed sensitive financial data in MSSQL with validation and audit controls. Supported microservices architecture and Docker-based containerization for scalability and reliability.
- Collaborated with teams to ensure system security and performance. Secured 10+ backend services per SDLC standards. Implemented secure multi-method payment processing. Built backend services in C#/.NET and frontend in Angular, adhering to secure coding practices.
- Developed and protected REST APIs to ensure safe, reliable transaction flows across finance systems.
- Worked with Microsoft SQL Server (MSSQL) to manage sensitive financial data, ensuring accuracy, consistency, and auditability.
- Supported microservices-based architecture and containerized services using Docker for scalability and reliability.
- Collaborated with finance, QA, and platform teams to maintain high availability, performance, and security of toll and payment systems.
- Developed and secured 10+ backend services with C#/.NET and Angular frontend components, following strict secure coding standards.



Charleston Area Alliance

West Virginia, United states

Security Analytics Engineer

01/2025 - 06/2025

Charleston Area Alliance, a non-profit economic group, boosts investment, supports businesses, and fosters growth in Charleston, WV.

- Google Cybersecurity Program: Developed secure APIs for sensitive data, aiding timely diagnosis. Assessed cybersecurity for 10+ businesses, evaluating infrastructure and data flows.
- Identified vulnerabilities, misconfigurations, and security gaps across servers, databases, APIs, and cloud environments.
- Conducted vulnerability assessments and security reviews, prioritizing risks and providing actionable remediation guidance.
- Designed secure APIs, ensuring access control, data integrity, and confidentiality. Used Python/SQL for anomaly detection. Recommended security tools, controls, and best practices to enhance infrastructure and application security.
- Collaborated with engineering and business stakeholders to ensure secure system design, compliance, and resilience.
- Developed security analytics to enhance risk visibility. Created Power BI dashboards for governance, access patterns, and suspicious behavior metrics.
- Analyzed large datasets using Python and SQL to identify trends in security posture, vulnerabilities, and operational risk.
- Developed secure application components using C#/.NET, supporting controlled data ingestion, validation, and access.
- Supported vulnerability management and data governance initiatives, enabling data-driven prioritization of security improvements.
- Conducted Google-organized security assessments, offering mitigation advice. Ensured secure data flow and compliance with security, engineering, and healthcare teams.



West Virginia State University

West virginia, united states

Graduate Research Assistant – Threat Detection & Research

08/2023 - 01/2025

West Virginia State University is a public research hub for cybersecurity, AI, and infrastructure security with industry partners.

- Optimized database performance by implementing efficient indexing, reducing query response time by 25%.
- Integrated UI components with secure backend systems. Conducted code reviews for quality and security. Authored 17 cybersecurity papers, publishing in IEEE and Springer. Developed ML/DL models for threat detection, e.g., SQL injection, insider threats, phishing, and ransomware.
- Conducted applied research on threat detection, IDS, and anomaly detection using real-world attack scenarios.
- Analyzed attack and defense characteristics across critical infrastructure, IoT, cloud platforms, and ICS/SCADA systems.
- Presented research at international cybersecurity conferences (IEEE Las Vegas 2023; Springer Nature Las Vegas 2024, San Francisco 2024).
- Contributed to research initiatives funded by Google, American Water, and Kanawha County, aligned with real-world security challenges.

Experience



Capgemini

Software Engineer

Bangalore India

05/2021 - 08/2023

Länsförsäkringar Bank (Sweden) is a leading **financial services and insurance provider**, offering secure banking, insurance, and digital financial solutions to customers across Sweden.

- Designed full-stack .NET applications for fraud detection in banking and insurance sectors, reducing false claims by 40%. Integrated APIs with real-time financial services, developed Power BI dashboards, and resolved production tickets for fraud and data issues.
- Designed and developed full-stack .NET applications for fraud detection in banking and insurance systems, reducing false claims by 40%.
- Integrated financial APIs, implemented security controls (input validation, access control, transaction integrity). Resolved incidents on fraud, data issues, unauthorized access. Collaborated with teams to enhance risk detection, monitoring, and fraud mitigation.



Appstix Technologies Private Limited

Software Analyst

Bangalore India

05/2019 - 05/2021

Technology services company providing software development and database solutions for enterprise applications. The company focuses on building scalable, reliable, and secure systems to support business operations.

- Developed SQL databases, Power BI dashboards, and data governance initiatives. Maintained secure SQL databases for enterprise applications. Ensured data integrity, access control, and secure coding.
- Enhanced MSSQL performance and addressed security vulnerabilities. Conducted assessments and supported remediation of database and application security issues.
- Collaborated with engineering teams to strengthen data protection, authentication, and secure transaction handling.
- Ensured data integrity, collaborated on secure coding. Resolved issues, optimized MS SQL databases. Addressed security risks and conducted vulnerability assessments.

Cybersecurity Research & Professional Development

 Machine Learning & Security Research Algorithms Used: Logistic Regression, SVM, Random Forest, LSTM, Bi-LSTM, Naive Bayes, Ensemble Learning, Anomaly Detection.	 Research Profiles & Publications 1. https://www.researchgate.net/profile/Venkatesh-Mannuru?ev=hdr_xprf 2. https://scholar.google.com/citations?authuser=4&user=0Hhq-0QAAAAJ	 Conferences & Book Publications Conferences: IEEE & Springer Nature (CA, NV, NC, DC), ACSE, CSCI
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Education

West Virginia State University Master of Science in Computer Science (Cybersecurity)	West Virginia, United states 2023 - 2025
Jawaharlal Nehru Technological University Anantapur Bachelor of Technology in Computer Science Engineering	India 05/2015 - 05/2019

Skills

Programming · C# · Java · Python · SQL · JavaScript · TypeScript · Burp Suite · Nessus · Metasploit · Nmap · OWASP ZAP · Kali Linux · Splunk (SIEM) · Wireshark · SQLMap · Secure Software Development · Application Security · Vulnerability Assessment · Penetration Testing · Security Analysis · Secure APIs · Authentication · Authorization · Threat Modeling · SQL Injection Prevention · Data Protection · Secure Financial Transactions · AWS · Azure · CI/CD Pipelines · Microservices-Based Architectures · DevSecOps · Cloud Security · FAANG Environments.